

Research paper

Advanced fetal cerebellar vermis segmentation and gestational age prediction in ultrasound imaging for prenatal neural development assessment

Qifeng Wang^{a,b,c}, Dan Zhao^d, Hao Ma^{a,b,c}, Bin Liu^{a,b,c,*}

^a Cancer Hospital of Dalian University of Technology, Dalian University of Technology, China

^b DUT School of Software Technology & DUT-RU International School of Information Science and Engineering, Dalian University of Technology, China

^c DUT-RU Co-Research Center of Advanced ICT for Active Life, Dalian University of Technology, China

^d Department of Ultrasound, Shengjing Hospital, China Medical University, China

ARTICLE INFO

Keywords:

Fetal ultrasound
Cerebellar vermis segmentation
Prenatal diagnostics
Fetal development stage

ABSTRACT

In prenatal diagnostics, accurate segmentation of the Cerebellar Vermis (CV) in fetal brain ultrasound images is essential for assessing fetal neural development. Traditional manual segmentation methods are prone to omissions and misdiagnoses, and even minor measurement errors can significantly affect fetal health and diagnostic accuracy. To address these challenges, we propose the Fetal Brain Feature Enhanced UNet (FB-FEUNet), an advanced segmentation network enhancing CV segmentation precision through specialized modules: the Week Embedding Module (WEM) for incorporating gestational timing, the Fusion Feature Attention Module (FFAM) for multisource feature integration, the Week Conditional Attention Module (WCAM) for gestational age-aware adjustments, and the Fusion Constraint Module (FCM) to enhance segmentation accuracy. The model was trained on the Fetal Brain Cerebellar Vermis Dataset (FB-CV), a curated collection of fetal brain ultrasound images designed to support robust evaluation. Experimental results demonstrate that FB-FEUNet achieves a Dice Coefficient of 0.8670 and an Intersection over Union of 0.7686, outperforming state-of-the-art methods in both accuracy and stability, while also providing faster inference times. These findings confirm the effectiveness of FB-FEUNet in addressing segmentation challenges and highlight its clinical potential to improve diagnostic accuracy, reduce manual errors, and improve fetal neural development assessments.

1. Introduction

In prenatal neurodevelopment assessment, accurate interpretation of fetal brain ultrasound images, especially the two-dimensional mid-sagittal plane images, is crucial for a comprehensive understanding of the fetal brain structure. During the critical gestational period of 18 to 32 weeks, monitoring the Cerebellar Vermis (CV) is vital, as its abnormal development can significantly impact newborn survival (Liu et al., 2009). As shown in Fig. 1, the positioning of the ultrasound probe relative to the mid-sagittal plane (FBMS Plane) highlights the sagittal structure of the fetal brain. However, the CV region often lacks clear boundaries in ultrasound images due to ultrasound limitations (e.g., speckle noise, artifacts, low contrast) and acoustic shadowing from the fetal skull, which obscure the region and complicate segmentation. Additionally, CV morphology varies across gestational stages and individual fetuses, further increasing boundary distinction challenges (Schmahmann and Sherman, 1998; Fatemi et al., 2012). Precise

CV segmentation enables the extraction of reliable biometric parameters — such as area, perimeter, length, and length-to-width ratio — which are routinely used in clinical assessments to evaluate neurodevelopmental progress (Torres et al., 2022). These measurements must align with gestational age norms to detect conditions such as Dandy–Walker malformation at an early stage. However, traditional measurement methods rely heavily on the subjective judgment of clinicians, introducing risks of variability and inaccuracy (Lipa et al., 2017). Even small deviations in measurement or gestational age matching can lead to diagnostic errors and compromise the assessment of fetal brain development (Zhao et al., 2016; Huang et al., 2018). To address these issues, we have proposed an automated CV segmentation method and developed a computer-aided diagnosis (CAD) system, with ongoing clinical trials (refer to Appendix A) aimed at providing a more objective and consistent assessment of fetal brain development.

* Corresponding author.

E-mail address: liubin@dlut.edu.cn (B. Liu).

<https://doi.org/10.1016/j.engappai.2025.113315>

Received 22 May 2024; Received in revised form 6 September 2025; Accepted 22 November 2025

Available online 25 November 2025

0952-1976/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

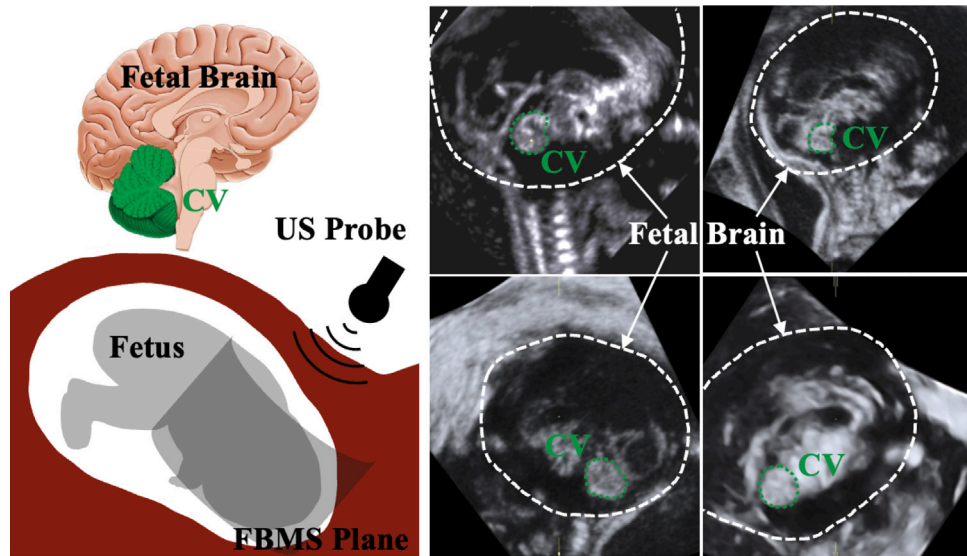


Fig. 1. Ultrasound Imaging of the Fetal Brain and Mid-Sagittal Plane. Left: Schematic of the sagittal plane structure and ultrasound probe positioning relative to the mid-sagittal plane (FBMS Plane). Right: Preprocessed FBMS planes from four fetuses, with the white region indicating valuable fetal brain information and the green region highlighting the Cerebellar Vermis (CV).

Our study addresses above limitations, offering an enhanced method for precise CV segmentation in fetal brain ultrasound imaging. This approach aims to improve diagnostic accuracy and efficiency, crucial for early detection of neurodevelopmental anomalies and refined fetal nervous system assessments. To effectively address these challenges, we developed the Fetal Brain Feature Enhanced UNet (FB-FEUNet), the contributions are as follows:

Week Embedding Module (WEM): Focuses on capturing time-related information pertinent to gestational age, providing the necessary context for the segmentation task. The WEM helps the model understand and adapt to changes in gestational age, enhancing segmentation accuracy.

Fusion Feature Attention Module (FFAM): Integrates texture, depth, and week number information, focusing on capturing key features in the fetal brain ultrasound images and strengthening the ability of network to recognize the CV structure.

Week Conditional Attention Module (WCAM): Utilizes week number information to fine-tune the attention mechanism of network, enabling better adaptation to structural changes in CV across different gestational stages.

Fusion Constraint Module (FCM): Combines the output layer and loss functions to provide comprehensive structural information about CV, aiding in more precise segmentation in complex areas.

Fetal Brain Cerebellar Vermis Dataset (FB-CV): We also developed a specialized dataset for fetal brain cerebellar vermis ultrasound images, FB-CV, comprising a large number of high-quality two-dimensional mid-sagittal plane ultrasound images and corresponding gestational week number information.

To improve the readability of the manuscript, the structure is organized as follows: Section 2 reviews related work on fetal brain ultrasound segmentation and gestational age analysis. Section 3 presents the proposed FB-FEUNet architecture and its key modules. Section 4 describes the construction and preprocessing of the FB-CV dataset. Section 5 details the experimental settings, including ablation and comparative studies. Section 6 discusses the clinical integration, potential extensions, current limitations, and future research directions.

2. Related work

2.1. Fetal ultrasound image segmentation

The advancements in deep learning have notably influenced medical image segmentation. The development of the U-Net model (Ronneberger et al., 2015) and its enhanced versions, such as UNet 3+ (Huang et al., 2020), along with transformer-based architectures like TransUNet (Chen et al., 2024) and Swin-UNet (Cao et al., 2022), have set a new standard in the field. These improvements have paved the way for more specialized applications in fetal ultrasound image segmentation. Particularly in fetal ultrasound, the shift from traditional methods to deep learning has been remarkable. The integration of sophisticated computational methods, as seen in systems for fetal weight estimation by Yu et al. (2008), and comprehensive evaluations by Rueda et al. (2013), have marked significant progress. Wu et al.'s implementation of a cascading fully convolutional network (Wu et al., 2017), Sobhaninia et al.'s multi-task deep learning approach (Sobhaninia et al., 2019), and Zeng et al.'s depth-supervised V-Net (Zeng et al., 2021), have shown how deep learning can enhance segmentation accuracy and efficiency. The application of such advanced techniques in measuring fetal head circumference and in placental segmentation (Chenarlogh et al., 2022; Zeng et al., 2022; Zimmer et al., 2023) highlights the depth of transformation in fetal ultrasound image analysis. Despite significant advancements in medical image segmentation with deep learning techniques for fetal ultrasound imaging, these methods still struggle with complex backgrounds and blurry boundaries. Thus, we introduce FB-FEUNet to overcome these limitations by enhancing feature extraction and edge detection, aiming for more accurate and robust CV region segmentation.

2.2. Attention mechanisms in medical image processing

In the field of deep learning, the application of attention mechanisms has significantly advanced image segmentation techniques across various domains, including MRI, CT scans, Ultrasound and remote sensing images. Represented by the Attention U-Net introduced by Oktay et al. (2018), this technology significantly enhances the ability of model to capture details by focusing on specific areas of the image. Subsequently, various improved versions emerged, such as the

multi-resolution Attention UNet by Thomas et al. (2020), the RA-UNet by Jin et al. (2020), and the 3D Attention UNet by Islam et al. (2020) and Li et al. (2020). For brain tumor and liver CT image segmentation, respectively, further expanding the application scope of attention mechanisms. Guo et al.'s (Guo et al., 2021) Spatial Attention UNet, Cai et al.'s Ma-UNet (Cai and Wang, 2022), which combines multi-scale and attention mechanisms, and the Attention Res-UNet by Maji et al. (2022) along with the Multi-Attention UNet by Sun et al. (2022) for brain tumor and remote sensing image segmentation, respectively, demonstrate efficiency. Recent transformer-based studies, such as the ViT cross-attentional model by Khaniki et al. (2024), also underscore the value of selective attention mechanisms in improving the recognition of brain structure from medical images. Recent studies have explored strategies such as integrating handcrafted and learned features (Serajian et al., 2024) or applying deep learning across diverse imaging modalities (Safari et al., 2025), providing valuable inspiration for enhancing robustness and generalization. While attention mechanisms in deep learning models have significantly improved image segmentation, they still face challenges with detail capture in low-contrast images and efficiency in resource-constrained environments. FB-FEUNet addresses these issues with an innovative attention mechanism that enhances detail detection and optimizes computational efficiency for practical clinical applications.

2.3. Ultrasound-based gestational age prediction

Gestational age prediction plays a key role in safeguarding maternal and infant health by tracking fetal development, estimating delivery dates, and identifying early pregnancy complications. The initial use of routine ultrasound by Campbell et al. (1985) marked a significant advancement in this field. Subsequently, the introduction of machine learning for fetal brain ultrasound analysis by Namburete et al. (2015) represented a pioneering integration of this technology into prenatal diagnostics. This was further enhanced by the work of Papageorgiou et al. (2016) and Li et al. (2016), who improved the accuracy of late gestational age and small-for-gestational-age infant risk assessment through machine learning and ultrasound. Wilson et al. (2016) expanded the data sources for more refined predictions, showcasing the trend towards utilizing diverse data for accuracy. Recent advancements, such as the method developed by Lu et al. (2020) for estimating fetal weight, and the approach by Wu et al. (2021) for detecting gestational diabetes early in pregnancy through machine learning, underscore the significant role of technology in enhancing prenatal care. Additionally, the application of machine learning by Naimi et al. (2018), Akhtar et al. (2020), and Fung et al. (2020) for fetal growth prediction, including the identification of large-for-gestational-age infants, further demonstrated the broadening scope of machine learning in gestational age estimation.

3. Proposed method

3.1. Overview

The FB-FEUNet, tailored for fetal brain ultrasound imaging, effectively segments the cerebellar vermis (CV). As presented in Fig. 2, its architecture includes four essential modules: (1) The Week Embedding Module (WEM) processes gestational age-related data; (2) The Fusion Feature Attention Module (FFAM) integrates texture and depth information with weekly data; (3) The Week Conditional Attention Module (WCAM) adapts to gestational changes; and (4) The Fusion Constraints Module (FCM) employs multiple loss functions for output optimization. This dual-input system, managing both image data I and gestational week information $week$, skillfully combines weekly data with embedded depth and texture features to refine CV segmentation. The module further enhances precision by accurately identifying smaller structures, crucial during gestational variations. By integrating heterogeneous features such as texture, semantic information, and gestational cues through above modules, the network is designed to better handle fetal position changes and noise variations.

3.2. Week Embedding Module (WEM)

As shown in the purple region of Fig. 2, this module is critical for analyzing fetal brain images across different gestational weeks, a key step in accurately assessing brain structure and development. We employ a dimensional encoding strategy that maps gestational weeks 18 to 32 into a 32-dimensional vector space. Each gestational week is assigned a unique vector F_{week} , capturing subtle differences and continuity between stages:

$$WEM(week) = \text{EmbeddingLayer}(week) = F_{week} \quad (1)$$

where $week$ denotes the input gestational week.

This mapping is achieved by a trainable weight matrix W_e :

$$F_{week} = W_e[week - 18] \quad (2)$$

Here, W_e has dimensions 16×32 , covering 15 specific gestational weeks plus an additional category for missing data ($W_e[16]$). Comparative studies on embeddings of 16, 32, 64, and 128 dimensions showed that the 32-D space strikes the best balance between representation quality, computational efficiency, and generalization. Lower dimensions lacked capacity, while higher dimensions incurred overfitting and extra overhead.

To accommodate cases where gestational week information is missing, the module incorporates a designated 'NULL' vector. This enables the network to make predictions based solely on image-derived features in such scenarios. By encoding gestational age as an explicit input, WEM provides auxiliary developmental context that supports fetal brain structure analysis, particularly when week information is available. This design improves the model's flexibility in handling heterogeneous data completeness while maintaining consistent input structure.

3.3. Fusion Feature Attention Module (FFAM)

As shown in the pink region of Fig. 2, this module integrates multi-dimensional feature streams, enhancing CV segmentation. Texture features from the Gray Level Co-occurrence Matrix (GLCM) (Partio et al., 2002), deep features from ResNet (Targ et al., 2016), and week number information are effectively fused, simplifying features post-fusion and improving CV segmentation accuracy. The implementation details are presented in Fig. 3, and the overall pseudocode for this process is as follows:

Algorithm 1 Fusion Feature Attention Module (FFAM)

Input:

Global Feature Map $g(C_g, W, H)$

Local Feature Map $l(C_l, W, H)$

Week Information $w(C_w)$

Steps:

1-1 (TFAM): Extract Texture Feature $F_{texture}(C_{texture}, W, H)$ by GLCM

1-2 (DFAM): Extract Deep Feature $F_{deep}(C_{deep}, W, H)$ by ResNet18

1-3 (WENM): Extract Week Feature $F_{week}(C_{week}, W, H)$ by WENM and FC

2 (Feature Fusion): Sum $F_{texture}, F_{deep}, F_{week}$

3 (Dimension Reduction): Apply Linear layer $\rightarrow (C_{reduced}, W, H)$

4 (Channel Adjustment): Use $1 \times 1 \text{ Conv} \rightarrow (C_{adjusted}, W, H)$

5 (Activation): Apply $\text{ReLU} \rightarrow 1 \times 1 \text{ Conv} \rightarrow \text{Sigmoid} \rightarrow \Psi(1, W, H)$

6 (Feature Refinement): Multiply Ψ with Input $l \rightarrow l'(C_l, W, H)$

Output:

Final Fusion Feature Map $l'(C_l, W, H)$

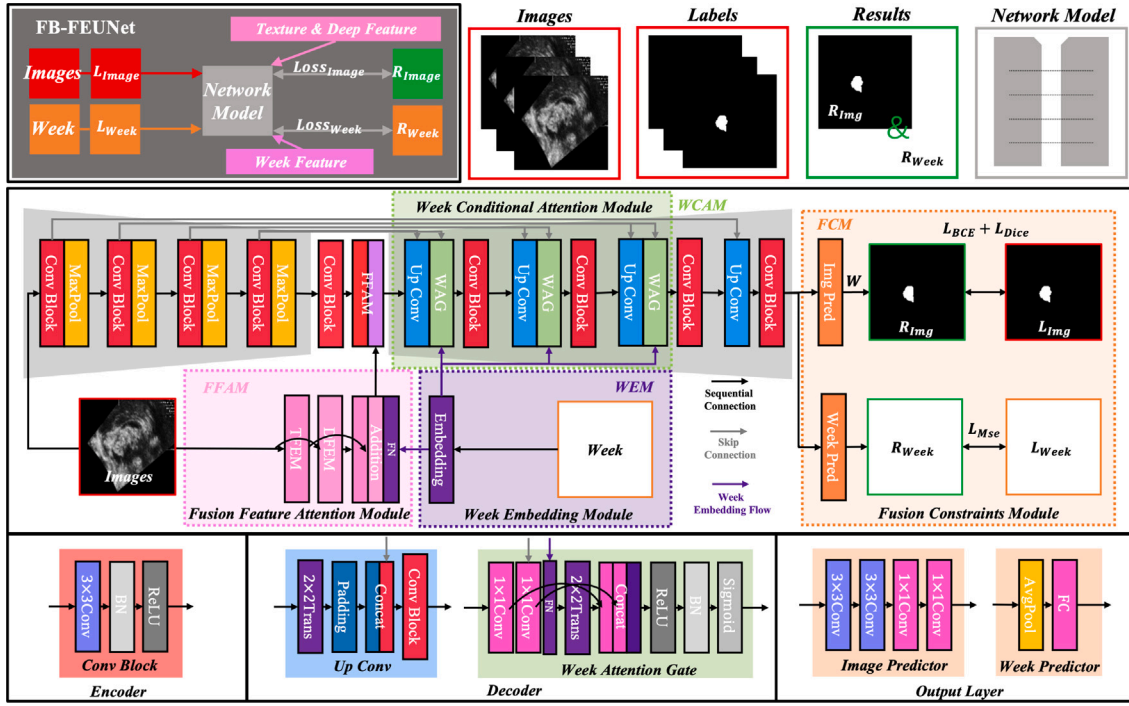


Fig. 2. Overall Network Architecture. The network comprises four main components: (1) Week Embedding Module (WEM) (2) Fusion Feature Attention Module (FFAM) (3) Week Conditional Attention Module (WCAM), and (4) Fusion Constraint Module (FCM). The network image input is represented in Red, including original images and labels; the input of week information is represented in Orange; the network prediction output is indicated in Green, also encompassing the segmentation results and week predictions; the baseline UNet is shown in Gray.

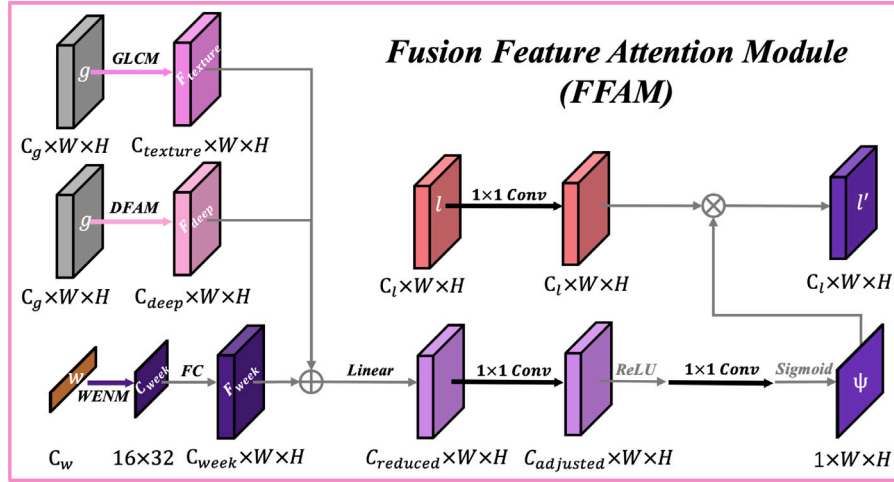


Fig. 3. Details of the Fusion Feature Attention Module (FFAM). The variables are defined as follows: C (Number of Channels), H, W (Feature Map Spatial Dimensions), g (Global Feature Map), l (Local Feature Map), w (Week Information), Ψ (Fusion Attention Map), and l' (Final Fusion Feature Map).

Step 1-1 Texture Feature Attention Module (TFAM): To extract statistical texture features, we employ GLCM to compute a co-occurrence matrix M for pixel pairs at a given distance d and direction θ in a grayscale image I . Typically, θ includes $0^\circ, 45^\circ, 90^\circ$, and 135° to capture texture orientations. From M , key texture features such as contrast (C), homogeneity (H), energy (E), and correlation ($Corr$) are derived. These measures effectively represent texture patterns:

$$C = \sum_{i,j} (i-j)^2 M(i,j), \quad H = \sum_{i,j} \frac{M(i,j)}{1 + (i-j)^2} \quad (3)$$

$$E = \sum_{i,j} M(i,j)^2, \quad Corr = \sum_{i,j} \frac{(i-\mu_i)(j-\mu_j)M(i,j)}{\sigma_i \sigma_j} \quad (4)$$

The resulting feature vector $F = [C, H, E, Corr]$ is transformed into a texture feature map F_{texture} , where each texture measure is assigned to a separate feature channel. Although GLCM is a hand-crafted approach potentially influenced by ultrasound signal variations, it supplies explicit statistical information that highlights subtle boundaries, effectively complementing high-level learned features. By integrating these two feature types, our framework captures both fine-grained local details and broader semantic cues.

Step 1-2 Deep Feature Attention Module (DFAM): A pretrained ResNet18 model captures high-level image features. The input image passes through its layers, extracting deep semantic information. Global Average Pooling (GAP) and a fully connected layer transform the high-dimensional feature map into a fixed-size vector F_{deep} for subsequent

processing. ResNet18 was selected for deep feature extraction because of its excellent ability to capture high-level semantic information in images, pre-trained on the ImageNet dataset (Deng et al., 2009). Paired with GLCM-based texture features, this design achieves a synergistic balance between learned representations and explicit texture cues, enhancing fetal brain ultrasound analysis.

Step 1-3 Week Encoding Module (WENM): As described in Section 3.2, WENM is employed to obtain week features F_{week} . These features are expanded using a fully connected (FC) layer to match the spatial dimensions (C_{week}, W, H). This ensures seamless integration with texture and deep features, enabling effective fusion across spatial dimensions.

Step 2 Feature Fusion: The three features $F_{texture}$, F_{deep} , and F_{week} are combined through simple addition. This straightforward method allows for an efficient and effective fusion of information, ensuring that all types of features contribute uniformly across the same scale.

Step 3 Dimensionality Reduction: The combined feature is then dimensionally reduced using a Linear layer, resulting in ($C_{reduced}, W, H$). The purpose of this step is to compress the dimensionality of the fused features, making them more suitable for subsequent processing and analysis, while also reducing computational complexity.

Step 4 to 6 Channel Adjustment, Activation, and Feature Refinement: A 1×1 convolution is applied to the fused features, followed by ReLU and Sigmoid activations. The Sigmoid output yields the fusion attention map Ψ , which highlights spatially important regions. This attention map is then element-wise multiplied with the local feature map l to produce the final fusion feature output l' .

3.4. Week Conditional Attention Module (WCAM)

As shown in the green region of Fig. 2, this module works in conjunction with FFAM to achieve precise CV segmentation. Its distinctiveness arises from the integration of encoded week information with encoder skip connection features, specifically focusing on key CV regions.

Integration with Encoder Features (depicted in the green box in the Decoder of Fig. 2): Embedding week information, F_{week} , merges with features F_{global} from the skip connections (illustrated by the gray arrow lines in Fig. 2) and F_{local} at various stages of the decoder, forming integrated features F_{concat} . This process, involving the input from Week Attention Gate (WAG), ensures that the attention mechanism is directed by both local image features and gestational age, which is vital for adapting the segmentation process to the developmental stage of the CV. Conditional attention adjustment is executed through the following steps:

Step 1-1 and 1-2 Adjustment of Global and Local Features: Global g and local l features are adjusted via 1×1 convolution, resulting in F_{global} and F_{local} .

Step 1-3 and 2 Adjustment and Transformation of Week Embedding Information: The week embedding information is first dimensionally adjusted through a fully connected layer, then further transformed via a Transformer to match the feature map dimensions, yielding F'_{week} .

Step 3 Feature Fusion: Unlike the feature fusion via addition in FFAM, WAG and WCAM utilize concatenation to integrate diverse features such as week information, global, and local image features. Concatenation effectively retains and combines distinct information from each feature source, especially vital when dealing with heterogeneous features. This approach enables the model to leverage the unique information from each type of feature, enhancing its ability to recognize critical CV areas, particularly when addressing structural variations during different gestational stages. Concatenation supports the model in learning complex relationships between features, improving segmentation accuracy.

Step 4 to 7: Following concatenation, ReLU activation, Batch Normalization (BN), and Sigmoid activation are applied. The final attention map Ψ is multiplied by the local features l to produce the attention-weighted feature map l' .

Algorithm 2 Week Attention Gate (WAG)

Input:

Global Feature Map $g(C_g, W, H)$

Local Feature Map $l(C_l, W, H)$

Week Embedding Information $w(16 \times 32)$

Steps:

1-1: Adjust global feature F_{global} channels by 1×1 Conv $\rightarrow (C_{global}, W, H)$

1-2: Adjust Local feature F_{local} channels by 1×1 Conv $\rightarrow (C_{local}, W, H)$

1-3: Adjust Week Embedding information by FC $\rightarrow (C_{week}, W, H)$

2 (Transformer Encoding): Transform F_{week} using Transformer $\rightarrow (C'_{week}, W, H)$

3 (Feature Fusion): Concatenate $F_{global}, F_{local}, F'_{week}$

4 (Activation): Apply ReLU activation $\rightarrow (C_{adjusted}, W, H)$

5 (Normalization): Apply Batch Normalization (BN) $\rightarrow (C_{adjusted}, W, H)$

6 (Attention Generation): Apply Sigmoid Activation $\rightarrow \Psi(1, W, H)$

7 (Feature Refinement): Multiply Ψ with Input $l \rightarrow l'(C_l, W, H)$

Output:

Attention-weighted Feature Map $l'(C_l, W, H)$

Conditional Attention Adjustment: The WCAM, which features three WAGs, dynamically refocuses on the critical areas of the CV based on gestational age information, delivering adjusted features F_{Adjust} . This enhancement in CV recognition is particularly significant for addressing structural variations at different gestational stages.

$$F_{Adjust} = \text{WCAM}(\text{WAG-3}(\text{WAG-2}(\text{WAG-1}(F_{int})))) \quad (5)$$

By incorporating week-conditioned attention, the model is designed to adaptively emphasize anatomically relevant features according to different gestational stages. This mechanism complements the global feature fusion process and aims to enhance segmentation adaptability to age-related structural variations. The effectiveness of this strategy is evaluated and demonstrated through experimental results in Section 5.

3.5. Fusion Constraints Module (FCM)

As shown in the orange region of Fig. 2, this module multiples loss functions (Binary Cross-Entropy (BCE), Dice, and Mean Squared Error (MSE)) are synergized to optimize segmentation accuracy, particularly for the precise delineation of CV. To further enhance the adaptability of the model in different stages of training and gestational ages, FCM integrates dynamically adjustable hyperparameters (α , β , and γ) along with category weights w_k , allowing adaptive optimization for different gestational weeks. Unlike fixed hyperparameters, these dynamically adjusted parameters allow the model to self-adapt based on the relative contributions of different loss components throughout training. The overall loss formula is defined as:

$$Loss = Loss_{Image} + Loss_{Week} \quad (6)$$

The image loss, $Loss_{Image}$, combines BCE and Dice losses, with α and β serving as hyperparameters to adjust the weights of the BCE loss and Dice loss respectively:

$$Loss_{Image} = \alpha L_{BCE}(R_{img}, L_{img}) + \beta L_{Dice}(R_{img}, L_{img}) \quad (7)$$

α and β are not fixed, but dynamically adjusted at each training epoch to reflect the changing influence of each loss term:

$$\alpha = \frac{L_{BCE}}{L_{BCE} + L_{Dice} + \sum_{k=1}^K w_k L_{MSE}}, \quad \beta = \frac{L_{Dice}}{L_{BCE} + L_{Dice} + \sum_{k=1}^K w_k L_{MSE}} \quad (8)$$

The BCE Loss ensures pixel-level classification accuracy, while the Dice Loss maintains continuity and structural integrity, crucial in scenarios of data imbalance:

$$L_{BCE} = -\frac{1}{N} \sum_{i=1}^N [g_i \log(p_i) + (1 - g_i) \log(1 - p_i)] \quad (9)$$

$$L_{Dice} = 1 - \frac{2 \sum_{i=1}^{H \times W} p_i g_i}{\sum_{i=1}^{H \times W} p_i + \sum_{i=1}^{H \times W} g_i} \quad (10)$$

To address category imbalance in week number prediction, the week loss, $Loss_{Week}$, utilizes MSE adjusted by a hyperparameter γ and incorporates category weights:

$$Loss_{Week} = \gamma \sum_{k=1}^K w_k L_{MSE}(R_{Week_k}, L_{Week_k}) \quad (11)$$

γ is dynamically adjusted based on the relative impact of MSE loss, and w_k represents the weight for category k and is calculated as:

$$\gamma = \frac{\sum_{k=1}^K w_k L_{MSE}}{L_{BCE} + L_{Dice} + \sum_{k=1}^K w_k L_{MSE}}, \quad w_k = \frac{1}{\log(c + p_k)} \quad (12)$$

where p_k is the proportion of category k in the dataset, and $c = 1.02$ for smoothing. The MSE Loss, crucial for accurate gestational age prediction, is defined as:

$$L_{MSE} = \frac{1}{N} \sum_{i=1}^N (p_i - g_i)^2 \quad (13)$$

For data instances labeled as ‘NULL’, where gestational age information is unavailable, the corresponding category weight is set to $w_k = 0$ in the week number prediction loss. This design excludes these samples from gestational regression supervision while retaining their image data for segmentation learning. During multi-objective optimization, α and β are used to dynamically balance the contributions of Binary Cross-Entropy (BCE) and Dice losses. In the early training phase, pixel-wise classification dominates the learning process (α is larger), while Dice loss becomes increasingly important in later stages (β increases), promoting a coarse-to-fine optimization trajectory. The hyperparameter γ governs the influence of the gestational week regression loss. As the model’s week prediction becomes more reliable, a gradual increase in γ facilitates more accurate modeling of fetal developmental trends. To address class imbalance in gestational week distribution, a logarithmic inverse-frequency weighting is applied to each category, ensuring that underrepresented gestational ages receive sufficient learning attention.

4. The dataset

To enhance prenatal ultrasound detection and fetal brain image analysis, we have developed the Fetal Brain Cerebellar Vermis Dataset (FB-CV), a high-quality, open-source, and meticulously annotated dataset focusing on cerebellar vermis (CV). Uniquely, FB-CV is the first dataset to include gestational age categorization, specifically for fetal brain cerebellar vermis studies, filling a critical gap in existing resources. This dataset is designed to bridge the existing data gaps in this field, establishing a novel benchmark for the automated recognition and analysis of CV. Interested researchers may contact the corresponding author via email to request the FB-CV dataset for academic use, subject to confidentiality and ethical approval.

4.1. Dataset collection and annotation

The dataset, created by Shengjing Hospital of China Medical University (Shenyang), comprises two-dimensional fetal brain ultrasound images targeting the CV region. It includes 1336 images from healthy singleton pregnancies within 18–32 gestational weeks, specifically excluding cases with miscarriage, intrauterine fetal demise (IUFD), postpartum deaths, or congenital anomalies to enhance segmentation precision. Gestational ages were verified using the last menstrual period and crown–rump length (CRL) measurements (Liu et al., 2009). Images

were obtained with high-end ultrasound devices such as GE Voluson E8, E10, and Samsung WS80 A, using advanced 3D transducers (RAB 4–8, RM 6C, eM6C) (AIUM (American Institute of Ultrasound in Medicine), 2018). In clinical practice, 2D mid-sagittal plane images are chosen from three-dimensional scans for detailed CV analysis. Our dataset was meticulously annotated by five experienced ultrasound technicians who determined the crucial plane for CV region marking, employing a collaborative review method to set a gold standard for annotation accuracy. This carefully curated dataset aims to support advancements in early fetal brain condition diagnostics by providing a reliable resource for CV region analysis.

4.2. Dataset statistics

4.2.1. Data composition

As shown in Fig. 4(a), the 1336-image dataset includes contributions from GE Voluson E8 (486 images), GE Voluson E10 (529 images), and Samsung WS80 A (321 images), ensuring a broad representation of imaging conditions. The original ultrasound images exhibited significant heterogeneity in size (typically 800–1200 pixels in width and 600–800 pixels in height), color rendering, and contrast levels, due to differences in acquisition devices, probe types, and operator settings. To address the heterogeneity of image sources, all ultrasound images were converted to grayscale and uniformly resized to 128×128 pixels to ensure consistency in resolution and contrast. Speckle noise was suppressed using median filtering. Standard data augmentation techniques were applied to the training set, including random rotations ($\pm 15^\circ$), horizontal flipping, and brightness adjustment, in order to enhance model robustness against clinical variability. To ensure a balanced and reliable dataset, we randomly allocated the images into training, validation, and test sets with an 8:1:1 ratio, resulting in 1070 training images, 133 validation images, and 133 test images. Additionally, stratified partitioning was employed to maintain a proportional representation of different gestational age groups across the three subsets, as illustrated in Table 1. This ensures that each subset reflects the natural distribution of gestational ages, reducing the risk of bias and improving the performance on diverse clinical data.

4.2.2. Week distribution

The dataset includes 1336 fetal CV ultrasound images, covering gestational weeks 18 to 32. However, as shown in Fig. 4(b), the distribution is not uniform, with a significantly higher concentration of images from the mid-gestation period (24–28 weeks) compared to the earlier (18–20 weeks) and later (30–32 weeks) stages. This distribution aligns with standard clinical practice, where mid-gestation scans are prioritized due to their superior imaging quality and crucial developmental insights.

A small portion of the dataset consists of images labeled as ‘NULL’, indicating that the exact gestational week is unknown. Despite this annotation variability, all images fall within the 18 to 32 weeks range. To enhance dataset diversity and better capture real-world clinical scenarios, these ‘NULL’ images are included exclusively in the training set, ensuring that the model learns from a wide variety of samples. However, as illustrated in Table 1, to maintain strict control over model evaluation, no ‘NULL’ images are included in the validation or test sets. By structuring the dataset in this manner, we ensure that the model receives a comprehensive and representative training set while maintaining rigorous evaluation standards with clearly defined gestational ages in the validation and test sets. This strategy ultimately contributes to a more generalizable and clinically applicable CV segmentation model.

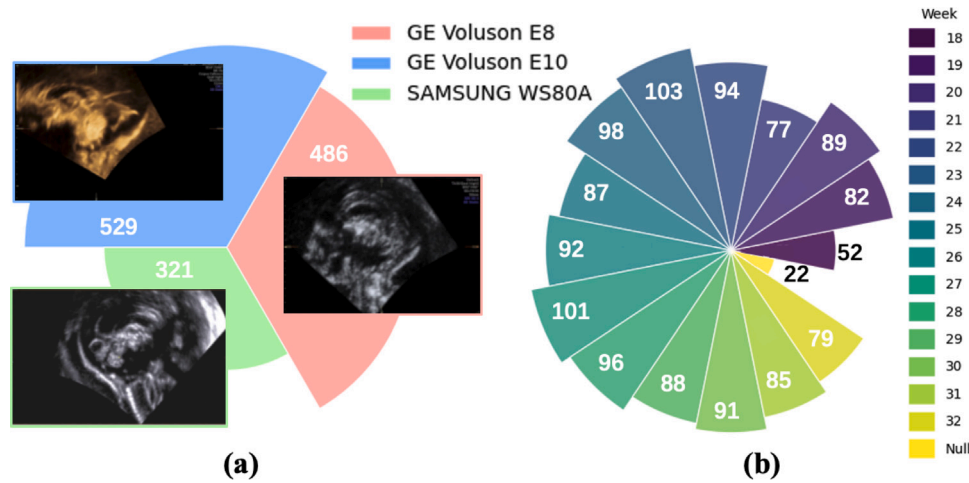


Fig. 4. Distribution of (a) Data Composition and (b) Week Distribution.

Table 1

Data distribution by gestational age group.

	Age	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	NULL
Total	1336	52	82	89	77	94	103	98	87	92	101	96	88	91	85	79	22
Training	1070	41	65	72	61	76	81	78	69	74	79	77	71	73	68	63	22
Validation	133	5	8	9	8	9	11	10	9	9	11	10	9	9	8	8	0
Testing	133	6	9	8	8	9	11	10	9	9	11	9	8	9	9	8	0

5. Experimental results

5.1. Implementation details

In this study, we used the dataset partitioning strategy described in Section 4, training with images of original size 128×128 . The model was developed using PyTorch 1.10.2 and executed on an Ubuntu workstation equipped with a 10-core 3.30 GHz CPU and two GeForce GTX 1080 Ti GPUs, with acceleration provided by CUDA 11.4. The training setup included 36 threads, with a batch size of 16. To ensure each model reaches its optimal performance, training continued until the validation performance stabilized, avoiding premature stopping or unnecessary overfitting. The initial learning rate was set at 0.001. Model optimization was carried out using the Adam optimization algorithm (Kingma and Ba, 2015), with momentum parameters β_1 and β_2 set at 0.9 and 0.99, respectively, and amsgrad enabled. Learning rate adjustments were managed by the 'ReduceLROnPlateau' scheduler (Bilogur, 2021), based on training losses or the Dice score of the validation set, with a reduction factor of 0.5 and a patience period of 10 epochs.

5.2. Results and analysis

To assess the performance of FB-FEUNet, we use metrics that include the Dice coefficient (Dice), MIoU, Precision (PRE), Recall (REC), Specificity (SPE), Accuracy (ACC), Hausdorff distance (HD), Mean Squared Error (MSE), and Weighted Mean Squared Error (Weighted MSE). These metrics, detailed in Appendix B Table 6, evaluate the precision in segmenting CV and predicting developmental periods in fetal brain ultrasound images.

5.2.1. Contrast experiment on FB-CV datasets

In our study, the FB-FEUNet network was benchmarked against advanced medical image segmentation methods, including UNet and its variants, for CV segmentation on the FB-CV dataset. To be fair, all the comparison methods in Table 2 use the same training methods and training parameters as ours, including the initial learning rate, batch size, the learning rate delay method and the data argument methods are

the same (including horizontal and vertical flip), also. As shown in Table 2, FB-FEUNet demonstrates superior performance, achieving a Dice coefficient of 86.70%, outperforming standard UNet by 8.91 points, and an MIoU of 76.86%, exceeding UNet by 10.92 points. It exhibits balanced precision (81.53%) and recall (91.94%), with specificity and accuracy surpassing 99.8%. The HD of 8.3797 mm suggests enhanced edge localization, and an inference time of 129.76 ms indicates efficiency. Additionally, the inclusion of Standard Deviation (SD) metrics for Dice and HD further confirms the robustness of the model. FB-FEUNet maintains a relatively low standard deviation (0.051 for Dice and 3.431 for HD), reflecting strong stability across different samples.

To complement Table 8 in Appendix D, we computed 95% confidence intervals (CIs) (Zou et al., 2004) for MIoU and HD using 133 test cases. FB-FEUNet achieved an MIoU of 0.7686 (95% CI: 0.757–0.780), slightly higher than MISSFormer (0.7662, CI: 0.754–0.777), and clearly above UNet (0.6694, CI: 0.654–0.684). For HD, FB-FEUNet reached 8.38 mm (CI: 7.74–9.02 mm), outperforming MISSFormer (8.59 mm, CI: 7.92–9.27 mm) and UNet (9.95 mm, CI: 9.18–10.72 mm). These CIs validate the consistency and reliability of our method's improvements across test samples.

In visual validation, Fig. 5 compares segmentation results and corresponding masks of three different samples. Columns l and m display the segmentation results of the CV region by FB-FEUNet and the ground truth, respectively, while columns b to k show the results of other methods. Most methods performed well in simple samples. However, as indicated by arrows and circles in Fig. 5, in challenging samples with extensive, complex CV targets, particularly in scenarios with blurred CV boundaries, indistinct foreground–background distinction, or noise interference, many methods showed noticeable segmentation errors. In contrast, FB-FEUNet excels, particularly evident in the second and third rows, where it provides more accurate segmentation with clearer outlines and precise details of the CV.

5.2.2. Application in computer aided diagnosis

In our study, to ensure experimental fairness and objective evaluation, we randomly selected 30 images from the test set for assessment. To evaluate the performance of FB-FEUNet, we enlisted five

Table 2

Performance of FB-FEUNet and Other Segmentation Methods on FB-CV dataset.

Methods	Dice (SD)	MIoU	PRE	REC	SPE	ACC	HD (SD)	Time(ms)
UNet (Ronneberger et al., 2015)	0.7749 (0.074)	0.6694	0.7473	0.8395	0.9980	0.9965	9.9473 (4.536)	102.36
ResUNet (Zhang et al., 2018)	0.8457 (0.093)	0.7470	0.8043	0.9077	0.9983	0.9974	8.5798 (3.877)	123.90
Attention UNet (Oktay et al., 2018)	0.8593 (0.047)	0.7598	0.8087	0.9333	0.9982	0.9976	8.3895 (3.652)	141.88
Channel UNet (Chen et al., 2019)	0.8576 (0.053)	0.7598	0.8106	0.9315	0.9982	0.9975	8.4184 (3.925)	138.57
KiUNet (Valanarasu et al., 2020)	0.8577 (0.058)	0.7596	0.7941	0.8727	0.9983	0.9970	9.2639 (4.220)	148.69
TransUNet (Chen et al., 2024)	0.8049 (0.089)	0.6980	0.7629	0.8729	0.9978	0.9966	9.6570(4.812)	193.17
MedT (Valanarasu et al., 2021)	0.7848 (0.075)	0.6711	0.7446	0.8661	0.9974	0.9962	10.2666 (5.039)	356.89
MISSFormer (Huang et al., 2022)	0.8607 (0.049)	0.7662	0.8112	0.9206	0.9983	0.9975	8.5934 (3.976)	159.78
Swin-Unet (Cao et al., 2022)	0.8598 (0.065)	0.7608	0.8038	0.8995	0.9981	0.9972	9.0327 (3.997)	203.44
SAM (Huang et al., 2023)	0.8622 (0.062)	0.7641	0.8129	0.9187	0.9982	0.9976	8.6065 (3.604)	138.61
FB-FEUNet(Ours)	0.8670 (0.051)	0.7686	0.8153	0.9194	0.9983	0.9976	8.3797 (3.431)	129.76

Precision evaluation methods referenced from Appendix 1.

Red indicates the best value, and Orange indicates the second-best value.

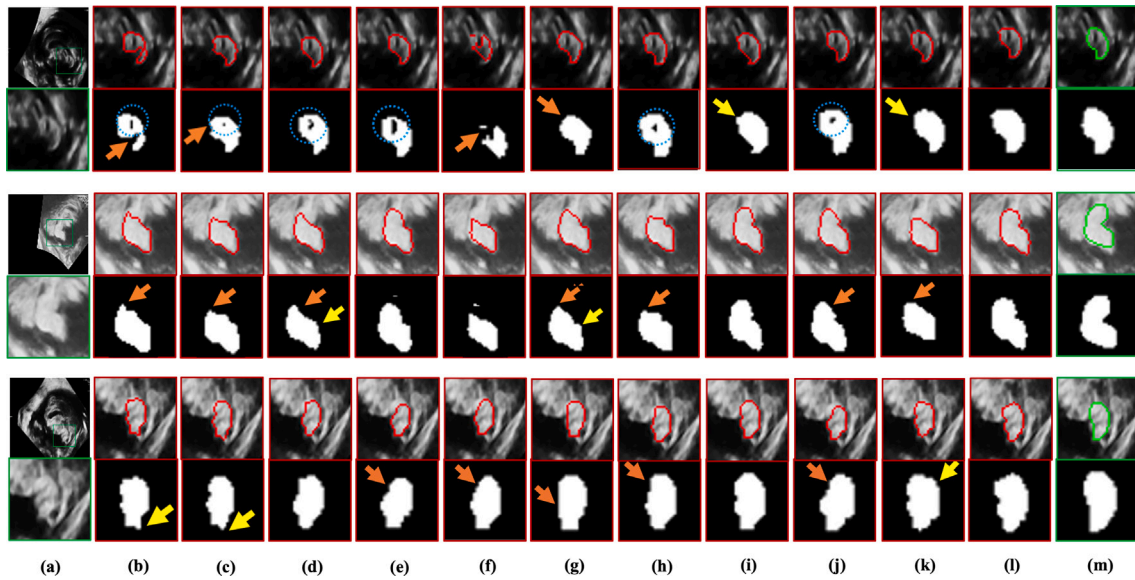


Fig. 5. Qualitative visual results on the FB-CV dataset compared with other methods in CV segmentation task. (a) Original images; (b) UNet (Ronneberger et al., 2015); (c) Res UNet (Zhang et al., 2018); (d) Attention UNet (Oktay et al., 2018); (e) Channel UNet (Chen et al., 2019); (f) KiUNet (Valanarasu et al., 2020); (g) TransUNet (Chen et al., 2024); (h) MedT (Valanarasu et al., 2021); (i) MISSFormer (Huang et al., 2022); (j) Swin-Unet (Cao et al., 2022); (k) SAM (Huang et al., 2023); (l) FB-FEUNet (Ours); (m) Ground truth. Orange arrow (Missing-segmentation); Yellow arrow (Over-segmentation); Blue circle (Non-contiguous errors). Our method can segment relatively complete CV regions and has a stronger ability to recognize details.

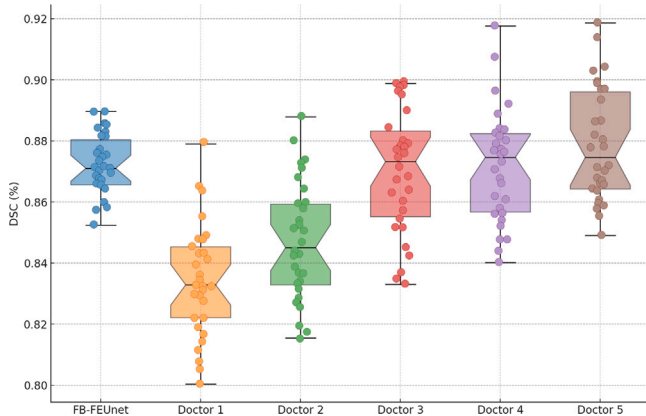


Fig. 6. Comparison of DSC and MSE of the FB-FEUNet (ours) with five ultrasound specialists in segmenting CV on the FB-CV dataset.

experienced sonographers from Shengjing Hospital of China Medical University, who were independent of the training dataset annotators. Each with over ten years of experience, these professionals independently annotated images and compared their work against the gold standard of our dataset. This procedure was designed to minimize potential biases and guarantee the broad reliability and validity of our evaluation results. As shown in Fig. 6, FB-FEUNet achieves a Dice coefficient (DSC) of 86.70%, outperforming experienced sonographers (85.94%), with a lower minimum mean squared error (MSE) (4.82% vs. 9.26%), demonstrating higher precision and stability. Although the improvements over experienced clinicians are not large in absolute value, they are meaningful in reducing inter-observer variability and ensuring consistency across different cases, especially in routine clinical workflows. The higher median, narrower interquartile range in the boxplot, and scatter plot distribution confirm its reliability across various scenarios, surpassing manual annotation consistency. While experienced clinicians achieve high accuracy, inter-observer variability persists, and less-experienced practitioners may introduce greater errors or struggle with segmentation. These results highlight the clinical value of FB-FEUNet in automated fetal brain ultrasound segmentation, reducing human errors, assisting clinicians, and enhancing diagnostic consistency.

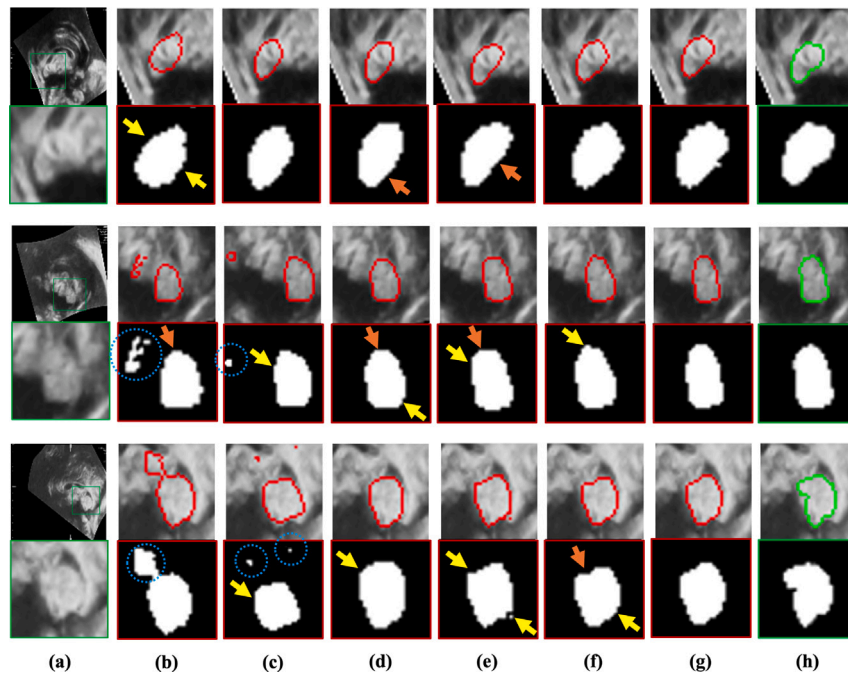


Fig. 7. Visual comparison of cerebellar vermis segmentation results among different configurations: (a) Original images; (b) UNet; (c) UNet+WEM; (d) UNet+WEM+WCAM; (e) UNet+WEM+WCAM+TFAM; (f) UNet+WEM+WCAM+DFAM; (g) UNet+WEM+WCAM+FFAM(TFAM+DFAM)(Ours); (h) Ground truth. **Orange arrow** (Missing-segmentation); **Yellow arrow** (Over-segmentation); **Blue circle** (Non-contiguous errors).

Table 3

Group statistics for developmental period prediction on the FB-CV dataset.

Exp	Corr	≤ 1	> 1	ACCW	MSE	wMSE
1	122	8	3	0.9173	0.150	0.331
2	126	6	1	0.9474	0.060	0.075
3	123	8	2	0.9248	0.105	0.180
4	123	7	3	0.9248	0.143	0.323

Corr (Exact match).

≤ 1 (Deviation within one week), > 1 (Deviation beyond one week).

5.2.3. Gestational age (week) prediction

The week prediction ability of FB-FEUNet was evaluated by cross-validation method. The data sets were randomized into training sets, validation sets, and test sets at a ratio of 8:1:1, and gestational age diversity was ensured for each set. Four experiments were conducted to improve the robustness of the evaluation. Specifically, to assess the accuracy of the model in predicting gestational weeks, week number labels for images in the test set were set to 'NULL'. Prediction outcomes were categorized into three types: exact accuracy, deviation within one weeks (considered minor error), and deviation beyond one weeks (considered major error). The performance of our model was assessed using Accuracy of Week Prediction (ACCW), Mean Squared Error (MSE), and Weighted Mean Squared Error (wMSE specifically designed to penalize larger deviations more severely, offering a more nuanced reflection of the precision of our model), as detailed in [Appendix B Table 6](#).

[Table 3](#) shows that the Accuracy for all groups is above 90%, indicating high precision in predicting fetal developmental weeks. Despite incorporating different error levels, the Weighted MSE remains low, highlighting robustness against various prediction errors. These results validate the accuracy in CV segmentation and the capability in predicting fetal development weeks, confirming the potential of this method in computer-aided diagnosis, especially for precise segmentation and developmental indicator prediction.

5.3. Ablation study

5.3.1. Impact of each component

[Table 4](#) and [Fig. 7](#) collectively illustrate the progressive improvements in CV segmentation through the integration of different modules. Beginning with the UNet baseline (a), the addition of WEM and FCM (b) enhances temporal awareness by leveraging gestational age data, leading to improved segmentation accuracy. Incorporating WCAM (c) further refines boundary delineation by enabling better adaptation to fetal structural variability.

A detailed comparison between (d) and (e) highlights two complementary approaches for feature augmentation. Specifically, (d) integrates TFAM with GLCM-based texture descriptors, effectively enhancing local texture representation and achieving refined boundary recognition, yielding a Dice score of 83.24% and the second-best ACCW (83.0%). Meanwhile, (e) employs DFAM to extract deeper semantic features, improving segmentation accuracy with a Dice score of 83.99% and a lower Hausdorff Distance (HD = 8.9547 mm).

Ultimately, FFAM (f) unifies these two feature streams, combining texture, depth, and gestational information to create a more robust and comprehensive feature representation. Notably, despite integrating multiple modules, the model maintains a compact size (8.76M parameters) and efficient inference time (129.76 ms), adding only 11.6% complexity over UNet. This moderate increase in complexity is well justified by the corresponding gains in segmentation accuracy and robustness. This fusion enables the model to achieve the highest Dice score (86.7%) and the best ACCW (85.5%), further validating the effectiveness of integrating hand-crafted texture descriptors with deep-learned semantic features. These results highlight the importance of leveraging both explicit texture cues and high-level representations to achieve superior CV segmentation in fetal brain ultrasound analysis.

5.3.2. Impact of each loss function

[Table 5](#) illustrates the effectiveness of the FB-FEUNet model composite loss function, which merges Dice and BCE losses and is enhanced

Table 4

Contributions of various component modules in cerebellar vermis segmentation task on FB-CV dataset.

Models	Dice (SD)	MIoU	HD (SD)	ACCW	Time (ms)
(a) UNet	0.7749 (0.074)	0.6694	9.9473 (4.536)	—	102.36
(b) (a)+WEM+FCM	0.8170 (0.069)	0.7157	9.1396 (4.254)	0.6900	107.58
(c) (b)+WCAM	0.8277 (0.063)	0.7145	9.5367 (4.087)	0.7400	113.42
(d) (c)+TFAM	0.8324 (0.059)	0.7300	9.0565 (3.641)	0.8300	119.84
(e) (c)+DFAM	0.8399 (0.054)	0.7379	8.9547 (3.592)	0.8150	120.73
(f) (c)+FFAM(Ours)	0.8670 (0.051)	0.7686	8.3797 (3.431)	0.8550	129.76

The ‘—’ value indicates data not available or not applicable.

Red indicates the best value, and Orange indicates the second-best value.

Table 5

Contributions of various loss functions in cerebellar vermis segmentation task on FB-CV dataset.

Loss image			Loss week	Dice	MIoU	PRE	REC	SPE	ACC	HD
Dice + BCE	Dynamic	Attention weight	MSE loss							
*			*	0.8334	0.7315	0.7777	0.9059	0.9978	0.9968	9.3093
*	*		*	0.8438	0.7463	0.7904	0.9226	0.9981	0.9974	8.6040
*		*	*	0.8474	0.7504	0.7996	0.9168	0.9981	0.9973	8.6307
*	*	*	*	0.8670	0.7686	0.8153	0.9194	0.9983	0.9976	8.3797

Precision evaluation methods referenced from Appendix 1.

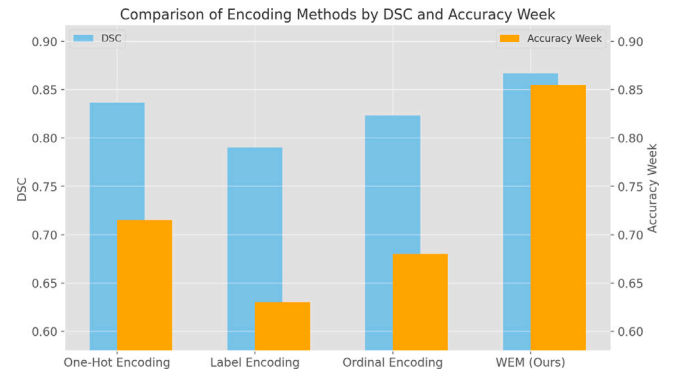
Red indicates the best value, and Orange indicates the second-best value.

* indicates inclusion of the corresponding loss function or strategy in the training configuration.

with a dynamic adjustment mechanism and attention weights. This approach adeptly incorporates gestational week features, markedly improving CV segmentation. The improvement in the Dice score from 0.8334 to 0.8670 underscores better recognition of CV continuity and integrity. Additionally, metrics such as MIoU, Precision, Recall, and Accuracy exhibit significant enhancements. Segmentation accuracy increases from 99.68% to 99.76%, and the reduction in Hausdorff distance from 9.3093 to 8.3797 indicates improved edge and detail handling in segmentation, especially in areas affected by gestational changes. Combining MSE loss for gestational week prediction with segmentation losses, the model excels in both segmentation and gestational age estimation. This integrated strategy ensures precise segmentation and accurate gestational week predictions, excelling in analyzing fine structural features of the CV. Thus, our method not only stands out in segmentation tasks but also brings innovation in loss function design, providing precise and robust outcomes in complex medical image analysis.

5.3.3. Impact of week encoding method

We evaluated the Week Embedding Module (WEM) against traditional encoding methods like One-Hot Encoding (Jason, 2020), Label Encoding (Anon, 2025), and Ordinal Encoding (Choong and Lee, 2017) for processing gestational week information in fetal brain imaging. Fig. 8 demonstrates that WEM outperforms these methods in both Dice Similarity Coefficient (DSC) and gestational week accuracy (Accuracy Week). WEM excels in capturing the continuity and subtleties of gestational age, which is crucial for predicting fetal brain development. Although CV segmentation is relatively insensitive to the method of gestational age encoding, the improvement in gestational week prediction emphasizes the importance of selecting the appropriate encoding technique for developmental analysis. In addition to its quantitative advantages, WEM introduces a trainable and biologically-informed representation that avoids the limitations of heuristic encodings. By modeling gestational weeks in a continuous embedding space, it enables the network to infer developmental trends more flexibly, even under noisy or incomplete annotations. Furthermore, the capability of WEM to handle missing data showcases its versatility and resilience in dealing with incomplete datasets.

**Fig. 8.** Evaluation of CV segmentation and developmental cycle prediction on the FB-CV dataset.

6. Conclusion

The Fetal Brain Feature Enhanced UNet (FB-FEUNet) advances fetal brain ultrasound analysis by integrating modules like the Week Embedding Module (WEM) and the Fusion Feature Attention Module (FFAM). Compared with purely large-scale, data-centric approaches, FB-FEUNet leverages specialized clinical insights — such as the importance of gestational weeks and ultrasound-specific texture cues — to achieve robust segmentation of the cerebellar vermis (CV), even when data volumes are relatively limited. Our experiments on the FB-CV dataset confirm that this custom architecture can effectively capture subtle fetal development details across varying gestational stages. In future work, we plan to enhance data diversity by incorporating images from multiple clinical sources and refining training strategies to better accommodate domain variability. Additionally, we will explore the integration of large-scale data augmentation techniques and broader imaging modalities, such as MRI or CT, to further improve model performance. Despite these future directions, our current results already demonstrate the practical utility of FB-FEUNet in real-world clinical settings, particularly where comprehensive datasets may be limited. We believe that the domain-specific enhancements not only

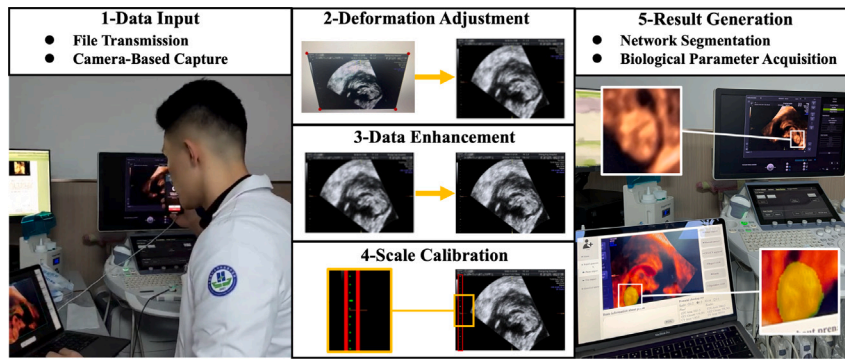


Fig. 9. The workflow of FB-FEUNet mobile APP.

enable more reliable diagnoses in such scenarios but also provide a complementary framework to standard architectures when extensive data is available, ultimately offering a balanced path toward improved fetal brain assessment across diverse medical imaging environments.

CRediT authorship contribution statement

Qifeng Wang: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Conceptualization. **Dan Zhao:** Resources, Funding acquisition, Data curation. **Hao Ma:** Software. **Bin Liu:** Writing – review & editing, Resources, Funding acquisition.

Ethical statement

All procedures carried out in studies involving human participants comply with the ethical standards of institutions and/or national research committees, as well as the 1964 Declaration of Helsinki and its subsequent amendments or similar ethical standards. All individuals involved in the study obtained informed consent. This study was approved by the Ethics Committee of Shengjing Hospital Affiliated to China Medical University with the ethics number 2022PS293K.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This study was supported by the National Natural Science Foundation of China (Nos. 62372079, 61972440 and 61572101), the Fundamental Research Funds for the Central Universities of China (No. DUT24YG129), the Natural Science Foundation of Liaoning Province of China (No. 2021-YGJC-23 and 2022-YGJC-73).

Appendix A

We integrated FB-FEUNet into a mobile app for clinical trials, streamlining the workflow into five steps: (1) image input via file or camera; (2) deformation correction; (3) noise reduction and contrast enhancement; (4) scale calibration for accurate measurements; and (5) automated segmentation and report generation, including area, perimeter, and length-to-width ratio. The real-time, end-to-end system fits seamlessly into obstetric workflows, with dual-input and lightweight design enabling use in low-resource settings. Future work will align predictions with expert consensus to build gestational age-specific reference curves for improved prenatal neurodevelopmental assessment (see Fig. 9).

Table 6

Metrics used for precision evaluation.

Metric	Formula
Dice Coefficient (DSC/Dice) ^a	$2 \times X \cap Y / (X + Y)$
Intersection over Union (MIoU) ^a	$ X \cap Y / (X \cup Y)$
Precision (PRE) ^b	$TP / (TP + FP)$
Recall (REC) ^b	$TP / (TP + FN)$
Specificity (SPE) ^b	$TN / (TN + FP)$
Accuracy (ACC) ^b	$(TP + TN) / (TP + TN + FP + FN)$
Hausdorff distance (HD) ^c	$\max(h(X, Y), h(Y, X))$
Accuracy of Week Prediction (ACCW) ^d	$\frac{N_T}{n}$
Mean Squared Error (MSE) ^e	$\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$
Weighted Mean Squared Error (wMSE) ^f	$\frac{1}{n} \sum_{i=1}^n w_i (Y_i - \hat{Y}_i)^2$

^a X (Prediction Region); Y (Ground Truth).

^b TP (True Positive); FP (False Positive); TN (True Negative); FN (False Negative).

^c $h(X, Y)$ (The max distance from a point in X to the closest point in Y); Unit: mm.

^d N_T (The number of corrects); n (The number of samples).

^e Y_i (The actual value for the i th sample.); $\hat{Y}_i$ (Predicted Value); n (The number of samples).

^f w_i (The weight for the i th sample error, with weights assigned based on the prediction error: 0 for no error, and 2 for errors greater than 1 week).

Appendix B

See Table 6.

Appendix C

See Table 7.

Appendix D

See Table 8.

Appendix E. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.engappai.2025.113315>.

Data availability

Data will be made available on request.

Table 7
List of acronyms.

Acronym	Full Name	Acronym	Full Name
FB-FEUNet	Fetal Brain Feature Enhanced UNet	FCM	Fusion Constraint Module
FB-CV	Fetal Brain Cerebellar Vermis Dataset	GLCM	Gray Level Co-occurrence Matrix
FBMS Plane	Fetal Brain Mid-Sagittal Plane	TFAM	Texture Feature Attention Module
WEM	Week Embedding Module	WAG	Week Attention Gate
FFAM	Fusion Feature Attention Module	DFAM	Deep Feature Attention Module
WCAM	Week Conditional Attention Module	WENM	Week Encoding Module

Table 8
Bootstrapped 95% Confidence Intervals for MIoU and HD.

Methods	MIoU	MIoU CI (95%)	HD	HD CI (95%)
UNet (Ronneberger et al., 2015)	0.6694	[0.654, 0.684]	9.95	[9.18, 10.72]
ResUNet (Zhang et al., 2018)	0.7470	[0.735, 0.759]	8.58	[7.92, 9.24]
Attention UNet (Oktay et al., 2018)	0.7598	[0.748, 0.772]	8.39	[7.77, 9.01]
Channel UNet (Chen et al., 2019)	0.7598	[0.748, 0.772]	8.42	[7.76, 9.08]
KiUNet (Valanarasu et al., 2020)	0.7596	[0.748, 0.771]	9.26	[8.54, 9.99]
TransUNet (Chen et al., 2024)	0.6980	[0.684, 0.712]	9.66	[8.84, 10.48]
MedT (Valanarasu et al., 2021)	0.6711	[0.656, 0.686]	10.27	[9.41, 11.09]
MISSFormer (Huang et al., 2022)	0.7662	[0.754, 0.777]	8.59	[7.92, 9.27]
Swin-Unet (Cao et al., 2022)	0.7608	[0.749, 0.772]	9.03	[8.36, 9.72]
SAM (Huang et al., 2023)	0.7641	[0.752, 0.776]	8.61	[7.99, 9.22]
FB-FEUNet (Ours)	0.7686	[0.757, 0.780]	8.38	[7.74, 9.02]

Red and orange still denote best and second-best mean values, respectively.
MIoU CIs obtained via non-parametric bootstrapping (1 000 resamples).
HD CIs computed as $\text{mean} \pm 1.96(\text{SD}/\sqrt{N})$, $N=133$.

References

AIUM (American Institute of Ultrasound in Medicine), 2018. AIUM practice parameter for the use of ultrasound to evaluate the female pelvis. *J. Ultrasound Med.* 37 (7), E13–E22.

Akhtar, F., Li, J., Azeem, M., Chen, S., Pan, H., Wang, Q., Yang, J.J., 2020. Effective large for gestational age prediction using machine learning techniques with monitoring biochemical indicators. *J. Supercomput.* 76, 6219–6237.

Anon, 2025. Label encoding in Python, GeeksforGeeks. [Online]. Available: <https://www.geeksforgeeks.org/ml-label-encoding-of-datasets-in-python/>.

Bilogur, A., 2021. LR schedulers, adaptive optimizers, in PyTorch training performance guide. [Online]. Available: <https://residentmario.github.io/pytorch-training-performance-guide/lr-sched-and-optim.html>.

Cai, Y., Wang, Y., 2022. MA-UNet: An improved version of unet based on multi-scale and attention mechanism for medical image segmentation. *ECNCT 2021*, In: Proc. Third Int. Conf. on Electronics and Communication; Network and Computer Technology, vol. 12167, SPIE, pp. 205–211.

Campbell, S., Warsof, S.L., Little, D., Cooper, D.J., 1985. Routine ultrasound screening for the prediction of gestational age. *Obs. Gynecol.* 65 (5), 613–620.

Cao, H., Wang, Y., Chen, J., Jiang, D., Zhang, X., Tian, Q., et al., 2022. Swin-unet: Unet-like pure transformer for medical image segmentation. In: *Computer Vision – ECCV 2022 Workshops*, vol. 13803, Springer, Cham, pp. 205–218.

Chen, J., Mei, J., Li, X., Lu, Y., Yu, Q., Wei, Q., et al., 2024. TransUNet: Rethinking the U-net architecture design for medical image segmentation through the lens of transformers. *Med. Image Anal.* 97, 103280. <http://dx.doi.org/10.1016/j.media.2024.103280>.

Chen, Y., Wang, K., Liao, X., Qian, Y., Heng, P.A., 2019. Channel-UNet: A spatial channel-wise convolutional neural network for liver and tumors segmentation. *Front. Genet.* 10, 1110.

Chenarlogh, V.A., Oghli, M.G., Shabanzadeh, A., Sirjani, N., Akhavan, A., Shiri, I., et al., 2022. Fast and accurate U-net model for fetal ultrasound image segmentation. *Ultrason. Imaging* 44 (1), 25–38.

Choong, A.C.H., Lee, N.K., 2017. Evaluation of convolutionary neural networks modeling of DNA sequences using ordinal versus one-hot encoding method. In: 2017 International Conference on Computer and Drone Applications. IConDA, IEEE, pp. 60–65.

Deng, J., Dong, W., Socher, R., Li, L.J., Li, K., Fei-Fei, L., 2009. ImageNet: A large-scale hierarchical image database. In: Proc. IEEE Conf. Computer Vision and Pattern Recognition. CVPR, IEEE, pp. 248–255.

Fatemi, S.H., Aldinger, K.A., Ashwood, P., Bauman, M.L., Blaha, C.D., Blatt, G.J., 2012. Consensus paper: pathological role of the cerebellum in autism. *Cerebellum* 11 (3), 777–807.

Fung, R., Villar, J., Dashti, A., Ismail, L.C., Staines-Urias, E., Ohuma, E.O., et al., 2020. Achieving accurate estimates of fetal gestational age and personalised predictions of fetal growth based on data from an international prospective cohort study: a population-based machine learning study. *Lancet Digit. Health* 2 (7), e368–e375.

Guo, C., Szemenyei, M., Yi, Y., Wang, W., Chen, B., Fan, C., 2021. Sa-unet: Spatial attention u-net for retinal vessel segmentation. In: 2020 25th International Conference on Pattern Recognition. ICPR, IEEE, pp. 1236–1242.

Huang, X., Deng, Z., Li, D., Yuan, X., Fu, Y., 2022. Missformer: An effective transformer for 2D medical image segmentation. *IEEE Trans. Med. Imaging*.

Huang, H., Lin, L., Tong, R., Hu, H., Zhang, Q., Iwamoto, Y., et al., 2020. Unet 3+: A full-scale connected unet for medical image segmentation. In: Proc. IEEE Int. Conf. Acoustics, Speech and Signal Process. ICASSP, pp. 1055–1059.

Huang, R., Namburete, A., Noble, A., 2018. Learning to segment key clinical anatomical structures in fetal neurosonography informed by a region-based descriptor. *J. Med. Imaging* 5 (1), 014007.

Huang, Y., Yang, X., Liu, L., Zhou, H., Chang, A., Zhou, X., et al., 2023. Segment anything model for medical images? *Med. Image Anal.* 103061.

Islam, M., Vibashan, V.S., Jose, V.J.M., Wijethilake, N., Utkarsh, U., Ren, H., 2020. Brain tumor segmentation and survival prediction using 3D attention unet, in brainlesion: Glioma, multiple sclerosis, stroke and traumatic brain injuries. In: 5th 5th Int. Workshop, BrainLes 2019. Springer, pp. 262–272.

Jason, B., 2020. Ordinal and one-hot encodings for categorical data, machine learning mastery. [Online]. Available: <https://machinelearningmastery.com/one-hot-encoding-for-categorical-data/>.

Jin, Q., Meng, Z., Sun, C., Cui, H., Su, R., 2020. RA-UNet: A hybrid deep attention-aware network to extract liver and tumor in CT scans. *Front. Bioeng. Biotechnol.* 8, 605132.

Khaniki, M.A.L., Mirzaebonehkhater, M., Manthouri, M., Hasani, E., 2024. Vision transformer with feature calibration and selective cross-attention for brain tumor classification. *Iran J. Comput. Sci.* 1–13.

Kingma, D.P., Ba, J., 2015. Adam: A method for stochastic optimization. In: Proc. Int. Conf. Learn. Represent. ICLR, pp. 1–15.

Li, J., Liu, L., Sun, J., Mo, H., Yang, J.J., Chen, S., et al., 2016. Comparison of different machine learning approaches to predict small for gestational age infants. *IEEE Trans. Big Data* 6 (2), 334–346.

Li, C., Tan, Y., Chen, W., Luo, X., Gao, Y., Jia, X., Wang, Z., 2020. Attention unet++: A nested attention-aware u-net for liver CT image segmentation. In: 2020 IEEE International Conference on Image Processing. ICIP, IEEE, pp. 345–349.

Lipa, M., Pooh, R.K., Wielgoś, M., 2017. Three-dimensional neurosonography—a novel field in fetal medicine. *Ginek. Polska* 88 (4), 215–221.

Liu, W., Cai, A.L., Li, T., Xin, Z.Q., Zhao, D., Yang, S., 2009. The value of three-dimensional ultrasound in the diagnosis of fetal cerebellar vermis hypoplasia. *J. China Ultrason.* 25 (1), 82–84.

Lu, Y., Fu, X., Chen, F., Wong, K.K., 2020. Prediction of fetal weight at varying gestational age in the absence of ultrasound examination using ensemble learning. *Artif. Intell. Med.* 102, 101748.

Maji, D., Sigedar, P., Singh, M., 2022. Attention res-unet with guided decoder for semantic segmentation of brain tumors. *Biomed. Signal Process. Control.* 71, 103077.

Naimi, A.I., Platt, R.W., Larkin, J.C., 2018. Machine learning for fetal growth prediction. *Epidemiol. (Cambridge Mass.)* 29 (2), 290.

Namburete, A.I., Stebbing, R.V., Kemp, B., Yaqub, M., Papageorgiou, A.T., Noble, J.A., 2015. Learning-based prediction of gestational age from ultrasound images of the fetal brain. *Med. Image Anal.* 21 (1), 72–86.

- Oktay, O., Schlemper, J., Folgoc, L.L., Lee, M.J., Heinrich, M.P., Misawa, K., et al., 2018. Attention U-net: Learning where to look for the pancreas. arXiv preprint, [Online]. Available: <https://arxiv.org/abs/1804.03999>.
- Papageorgiou, A.T., Kemp, B., Stones, W., Ohuma, E.O., Kennedy, S.H., Purwar, M., et al., 2016. Ultrasound-based gestational-age estimation in late pregnancy. *Ultrasound Obs. Gynecol.* 48 (6), 719–726.
- Partio, M., Cramariuc, B., Gabbouj, M., Visa, A., 2002. Rock texture retrieval using gray level co-occurrence matrix. In: *Proc. of 5th Nordic Signal Processing Symposium*, vol. 75.
- Ronneberger, O., Fischer, P., Brox, T., 2015. U-net: Convolutional networks for biomedical image segmentation. MICCAI, In: *Proc. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351, Springer, Cham, pp. 234–241.
- Rueda, S., Fathima, S., Knight, C.L., Yaqub, M., Papageorgiou, A.T., Rahmatullah, B., et al., 2013. Evaluation and comparison of current fetal ultrasound image segmentation methods for biometric measurements: a grand challenge. *IEEE Trans. Med. Imaging* 33 (4), 797–813.
- Safari, K., Rodriguez Vila, B., Pierce, D.M., 2025. Automated detection of microcracks within second harmonic generation images of cartilage using deep learning. *J. Orthop. Res.* 43 (6), 1101–1112.
- Schmahmann, J.D., Sherman, J.C., 1998. The cerebellum contributes to higher functions during development: evidence from a series of children surgically treated for posterior fossa tumours. *Brain* 123 (5), 1051–1061.
- Serajian, M., Marini, S., Alanko, J.N., Prosperi, N.R., Boucher, C., 2024. Scalable de novo classification of antibiotic resistance of mycobacterium tuberculosis. *Bioinformatics* 40 (suppl. 1), i39–i47.
- Sobhaninia, Z., Rafiei, S., Emami, A., Karimi, N., Najarian, K., Samavi, S., et al., 2019. Fetal ultrasound image segmentation for measuring biometric parameters using multi-task deep learning. In: *Proc. IEEE Eng. Med. Biol. Soc. EMBC*, pp. 6545–6548.
- Sun, Y., Bi, F., Gao, Y., Chen, L., Feng, S., 2022. A multi-attention unet for semantic segmentation in remote sensing images. *Symmetry* 14 (5), 906.
- Targ, S., Almeida, D., Lyman, K., 2016. ResNet in ResNet: Generalizing residual architectures. arXiv preprint, [Online]. Available: <https://arxiv.org/abs/1603.08029>.
- Thomas, E., Kumar, P.S.J., Kumar, S., Horo, A., Niyas, S., Vinayagamani, S., et al., 2020. Multi-res-attention UNet: a CNN model for the segmentation of focal cortical dysplasia lesions from magnetic resonance images. *IEEE J. Biomed. Health Inform.* 25 (5), 1724–1734.
- Torres, H.R., Morais, P., Oliveira, B., Birdir, C., Rüdiger, M., Fonseca, J.C., et al., 2022. A review of image processing methods for fetal head and brain analysis in ultrasound images. *Comput. Methods Programs Biomed.* 215, 106629.
- Valanarasu, J.M.J., Oza, P., Hacıhaliloglu, I., Patel, V.M., 2021. Medical transformer: Gated axial-attention for medical image segmentation. MICCAI, In: *Proc. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 12901, Springer, pp. 36–46.
- Valanarasu, J.M.J., Sindagi, V.A., Hacıhaliloglu, I., Patel, V.M., 2020. KiU-net: Towards accurate segmentation of biomedical images using over-complete representations. MICCAI, In: *Proc. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 12264, Springer, pp. 363–373.
- Wilson, K., Hawken, S., Potter, B.K., Chakraborty, P., Walker, M., Ducharme, R., et al., 2016. Accurate prediction of gestational age using newborn screening analyte data. *Am. J. Obstet. Gynecol.* 214 (4), 513.e1.
- Wu, L., Xin, Y., Li, S., Wang, T., Heng, P.A., Ni, D., 2017. Cascaded fully convolutional networks for automatic prenatal ultrasound image segmentation. In: *Proc. IEEE Int. Symp. Biomedical Imaging. ISBI*, pp. 663–666.
- Wu, Y.T., Zhang, C.J., Mol, B.W., Kawai, A., Li, C., Chen, L., et al., 2021. Early prediction of gestational diabetes mellitus in the Chinese population via advanced machine learning. *J. Clin. Endocrinol. Metab.* 106 (3), e1191–e1205.
- Yu, J., Wang, Y., Chen, P., 2008. Fetal ultrasound image segmentation system and its use in fetal weight estimation. *Med. Biol. Eng. Comput.* 46 (12), 1227–1237.
- Zeng, W., Luo, J., Cheng, J., Lu, Y., 2022. Efficient fetal ultrasound image segmentation for automatic head circumference measurement using a lightweight deep convolutional neural network. *Med. Phys.* 49 (8), 5081–5092.
- Zeng, Y., Tsui, P.H., Wu, W., Zhou, Z., Wu, S., 2021. Fetal ultrasound image segmentation for automatic head circumference biometry using deeply supervised attention-gated V-Net. *J. Digit. Imaging* 34, 134–148.
- Zhang, Z., Liu, Q., Wang, Y., 2018. Road extraction by deep residual U-net. *IEEE Geosci. Remote. Sens. Lett.* 15 (5), 749–753.
- Zhao, D., Cai, A.L., Zhang, J., Wang, B., Xie, L.M., 2016. Quantitative assessment of the fetal cerebellar vermis with three-dimensional ultrasound and MRI. *J. China Med. Imaging Technol.* 32 (1), 109–113.
- Zimmer, V.A., Gomez, A., Skelton, E., Wright, R., Wheeler, G., Deng, S., et al., 2023. Placenta segmentation in ultrasound imaging: Addressing sources of uncertainty and limited field-of-view. *Med. Image Anal.* 83, 102639.
- Zou, K.H., Warfield, S.K., Bharatha, A., Tempany, C.M.C., Kaus, M.R., Haker, S.J., et al., 2004. Statistical validation of image segmentation quality based on a spatial overlap index: scientific evaluation. *Academic Radiol.* 11 (2), 178–189.